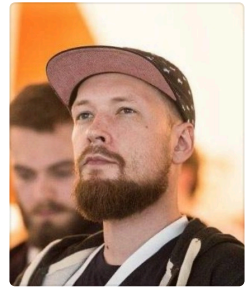


Sergey Rubanov

Engineering **AI agent infrastructure**, **web platform tooling** & **developer tools**, from specifications and engines to production systems.



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01 Profile

Software engineer and independent researcher working on AI agent infrastructure, the web platform, developer tooling, and open source.

Contributes to **TC39** and **W3C** work on JavaScript semantics, browser behavior, conformance tests, and developer-facing APIs.

02 Core Skills

AI / AGENT SYSTEMS

agent orchestration tool execution evaluation harnesses
provider routing OpenAI / Anthropic APIs llama.cpp MCP / ACP

WEB PLATFORM / STANDARDS

ECMA-262 ECMA-402 TC39 W3C Test262 WebAssembly
JavaScriptCore WebKit browsers/webviews

LANGUAGES

TypeScript Go C++ Zig Coq / Rocq

PLATFORM / OBSERVABILITY

OpenTelemetry traces/metrics/logs Prometheus Grafana
OTel Collector Docker Kubernetes Terraform AWS GCP
Cloudflare

03 Selected Projects

Hecate — AI WORKSPACE / RUNTIME

Local-first workspace for local and cloud AI models, project context, and supervised coding agents, with provider APIs, approvals, diff review, usage reporting, and OpenTelemetry.

Stack Go · TypeScript · React · Tauri · SQLite · OpenTelemetry
🌐 hecate.sh · 🐙 github.com/hecatehq/hecate

Cynic — ECMASCRIPT / WEBASSEMBLY ENGINE

Strict-only ECMAScript and WebAssembly engine in Zig with frozen primordials, hardened realms, host-controlled code generation, and no Annex B.

Stack Zig · ECMAScript · WebAssembly · Test262 · WebAssembly testsuite
🌐 sergey.works/cynic · 🐙 github.com/chicoxyzy/cynic

Pragmatist — REPRODUCIBLE VERIFICATION

Research tool for proving bugs in ECMAScript specifications, TypeScript, JavaScript engines, and WebAssembly runtimes using pinned source data, Coq/Rocq mechanization, differential runs, patch scaffolding, and deterministic replay.

Stack TypeScript · Coq/Rocq · ECMA-262 · WebAssembly · tsgo · Zod

TC39 MCP — MCP SERVER

MCP server for clause-level ECMA-262 and ECMA-402 search, cross-spec references, edition diffs, git history, Test262 and proposal lookup, and SHA-pinned local or hosted responses.

Stack TypeScript · ECMA-262 · ECMA-402 · Cloudflare Workers · VitePress · Zod
🌐 mcp.xyzyzylabs.ai/tc39/ · 🐙 github.com/xyzyzylabs/tc39-mcp

WASM MCP — MCP SERVER

Read-only MCP server for WebAssembly specifications with instruction and clause lookup, opcode and stack-signature metadata, proposal search, and SHA-pinned local or hosted responses.

Stack TypeScript · WebAssembly · Cloudflare Workers · VitePress · Zod
🌐 mcp.xyzyzylabs.ai/wasm/

Independent Software Engineer / Consultant

BARCELONA

Sep 2025 – Present

Independent engineering and research across AI agents, web standards, developer tools, and open source.

- Build local-first agent tooling for model access, supervised coding sessions, approvals, diff review, and telemetry.
- Develop standards and conformance tools for ECMAScript, TypeScript, and WebAssembly, including source pinning and differential testing.
- Contribute to TC39/ECMA-262 and Test262 by analyzing proposal trade-offs, refining specification text and semantics, and collaborating with browser vendors, JS engines, and the standards community.
- Consult on agent architecture, developer tooling, observability, and deployment.

Invited Expert • TC39

May 2019 – Present

- Co-authored the **Promise.any** proposal and contributed editorial improvements to ECMA-262.
- Participate in proposal discussions, evaluate compatibility and ergonomics trade-offs, and develop Test262 tests to validate JavaScript behavior across engines.

Software Engineer • Firetiger

May 2024 – May 2025

US • REMOTE

Firetiger acquired Dispatch.run four months into the role.

Firetiger OBSERVABILITY PLATFORM

- Built product features and infrastructure for an OpenTelemetry-based observability platform.
- Helped early customers instrument Python and TypeScript services and deploy integrations across Vercel, Kubernetes, and Terraform-managed infrastructure.

Stack Python • TypeScript • Node.js • Next.js • Vercel • Docker • Kubernetes • Terraform • AWS / GCP • OpenTelemetry • Prometheus • Grafana • OTel Collector

Dispatch.run DURABLE FUNCTIONS PLATFORM

- Owned the CLI workflow for local development, deployment, and handoff to cloud execution.
- Evaluated running Go workloads as WebAssembly on Cloudflare.

Stack Go • WebAssembly • Cloudflare

Software Engineer • Socket Supply Co

Mar 2022 – Jan 2024

US • REMOTE

- Developed core Socket Runtime components in C++ for execution, platform integration, and native capabilities.
- Expanded JavaScript APIs for Node.js compatibility across mobile and desktop hosts.
- Built tooling for project setup, local development, and runtime debugging.

Stack C++ / C / Objective-C++ • JavaScript / TypeScript • Node.js • WebKit- and Chromium-based webviews (WKWebView, libwebkit2gtk, WebView2)

Software Engineer • RealAdvisor

Dec 2019 – Sep 2021

GENEVA • REMOTE

- Developed a multi-tenant CRM and custom document editor with reusable layout templates.
- Optimized the document rendering pipeline to improve load times and editor responsiveness.

Stack React / Svelte • Flow • Node.js • Relay • Docker + Composer • Headless Chromium • GCP

Frontend Developer Lead · EXANTE

MALTA · REMOTE

Dec 2014 — Aug 2019

- Led frontend development of a web trading terminal, browser extension, Ledger hardware-wallet apps, and cryptocurrency products.
- Mentored and interviewed web developers while leading delivery across React/Preact, TypeScript, RxJS, D3, Redux/MobX, and browser-extension work.

Stack React/Preact · Redux/MobX · RxJS · D3 · TypeScript · C**Senior Developer · NPTV**

MOSCOW · RUSSIA

Oct 2013 — Nov 2014

- Built a JavaScript SDK for an interactive TV platform, similar to React Native; helped adopt SpiderMonkey and later migrate to V8.
- Maintained build tooling and DevTools extensions and prototyped client-side platform features.

Stack JavaScript · Babel · React · Chrome DevTools extensions**Frontend Team Lead · KM Media**

MOSCOW · RUSSIA

Sep 2012 — Jul 2013

- Led frontend delivery for an advertising platform.
- Introduced unit testing with Karma and Jasmine to improve regression coverage.
- Worked with backend and product teams on integration and data-quality issues.

Stack Angular · JavaScript**Senior Developer · Comfortel**

ORYOL · RUSSIA

Sep 2009 — Jul 2013

- Built a CRM and integrations with Genesys Agent Desktop and Oktell for call-center workflows.
- Promoted to Team Lead in 2012; led delivery of customer-interaction tracking and operator workflows.
- Built a web-based operator-script generator for call-center teams.
- Worked with MySQL, MSSQL, and Oracle databases for CRM data, reporting, and call-center integrations.

Stack PHP · JavaScript · MSSQL · MySQL · OracleDB**Master of Computer Science · Oryol State Technical University**

ORYOL · RUSSIA

2009

Languages — Russian (Native) · English (Proficient) · Spanish (Novice)